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TOS DESIGN OPTIMIZATION: PLANT DIVERSITY SAMPLING

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1 BACKGROUND AND GOALS

The National Ecological Observatory Network (NEON) generates data and samples to advance understanding of ecological processes and inform management of US ecosystems. Multiple tiers of requirements provide traceable links between this high-level NEON mission statement and the protocols that direct data collection. The Terrestrial Observation System (TOS) science design for plant diversity ([RD03]) and the resulting study design, protocol ([RD03]), and data product (Plant Presence and Percent Cover, DP1.10058.001) are designed to enable status and trends in composition, abundance, and richness of plant species and function types at multiple scales.

1.1 Description of sample design and available data

NEON's plot-based sampling of plant species diversity satisfies the requirements that "NEON's measurement strategy will include biological invaders and diversity of key organismal groups" and "NEON measurements of plant diversity will include native and invasive plant species (Schimel et al. 2011)." Species are observed with the Carolina Vegetation Survey (Peet et al. 1998) multi-scale plot design that borrows from a sampling approach pioneered by Robert Whittaker (Shmida 1984, Stohlgren 1995). NEON samples with a 20 x 20m square plot with four 10m x 10m subplots containing nested 10m² and 1m² subplots (Figure 1). Specifically:

- The identity of each species (or the lowest taxonomic rank possible) according to naming conventions maintained by the US Department of Agriculture, Natural Resources Conservation Service PLANTS database (USDA, NRCS 2016) is recorded in each subplot - eight 1m², eight 10m², four 100m².
- Estimates of taxon-specific abundance is made with estimates of cover within the 1m² subplots.

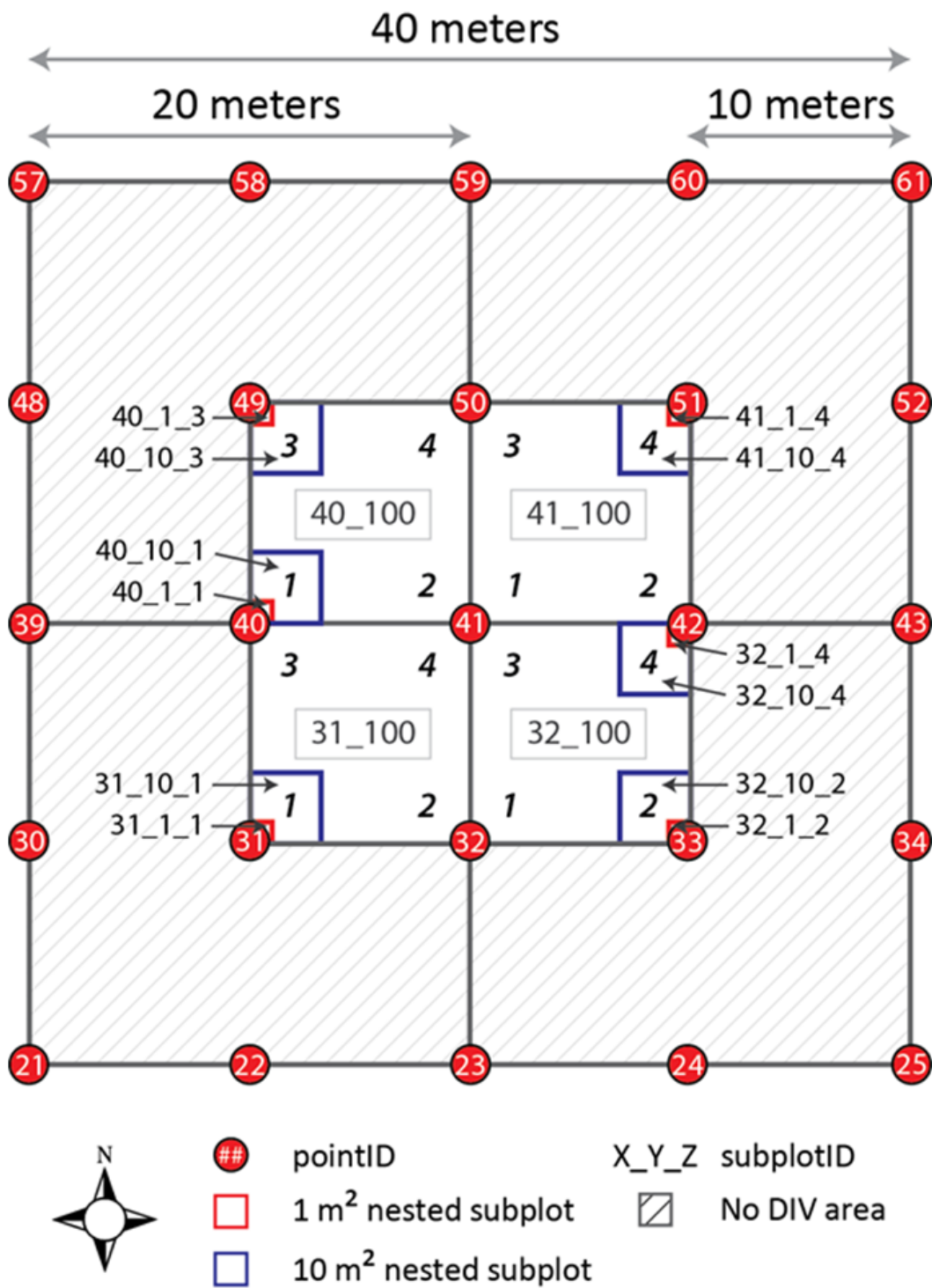


Figure 1: A diagram of the multiscale plot for observations of plant cover-abundance and occurrence data.

At each of 47 terrestrial NEON sites, the NEON spatial sampling design for organisms and soil directs observations to (Thorpe et al. 2016):

- Tower Base Plots: Randomly placed plots near the NEON flux tower designed to calibrate sensor-based measures of atmosphere and soil
- Distributed Base Plots: Plots to measure variability across larger landscapes according to a stratified-random design

The plant diversity protocol is implemented at three of the 20–30 Tower Base Plots. The majority of the plant species diversity sampling effort occurs at the Distributed Base Plots (Thorpe et al. 2016). Ecological literature (e.g., Stohlgren 2007), early input from the ecological community, and power analyses (Thompson et al. 2012, Barnett et al. 2018) resulted in an initial sample size of 20–30 Distributed Base plots at each site.

The frequency and timing of sampling is guided by NEON requirements (Schimel et al. 2011). Implementing the protocol at each plot at least one time each year enables the quantification of annual rates of change. These sampling bouts occur during a one to two month period when the majority of species flower or possess other parts conducive to identification. This sampling window targets annual peaks in greenness as measured by MODIS, and incorporates input from local ecologists and NEON phenology observations (Meier et al. 2023). The few NEON sites characterized by distinct bimodal patterns in peak greenness populated disparate suites of apparent species are sampled twice annually. For example, the Sonoran desert (represented by the NEON site at the Santa Rita Experimental Range) gets rain in late winter and a monsoonal moisture pulse late in the summer. Plant species adapted to the particular intersection of temperature and the timing and amount of precipitation grow and flower and can be identified only during one of the two distinct events (Ogle and Reynolds 2004).

For the purpose of these analyses, data includes all available Distributed Base Plots from all 47 NEON sites (NEON 2025) downloaded directly from the application programming interface (API) and stacked with 'neonUtilities::stackByTable()' function on 03/01/2026.

1.2 Analytical Goals

Statistically rigorous analyses are needed to assess the capacity of NEON data to address Observatory goals and to guide iterative sampling design optimization efforts. Initial spatial and temporal sampling designs for the NEON Terrestrial Observation System (TOS) and Aquatic Observation System (AOS) were developed in collaboration with Technical Working Groups (TWGs) comprised of community experts, and were captured in Science Design documents (RD[01]). The initial designs relied on analysis of published datasets (where relevant), analysis of NEON prototype data collection efforts, and subject matter expertise.

As the NEON Observatory matures, it is critical for initial design assumptions to be tested with multiple years of data collected from NEON sites. Analyses must:

- Describe where the data support reduction of sampling effort without compromising NEON's ability to meet design requirements (e.g., target sites for reduction in sampling effort, spatial scales where replication is not necessary, etc.).

- Evaluate effects of design modifications on uncertainty and ability to detect changes for key response variables (e.g., effects of reduced spatial and/or temporal sampling effort).

Analysis and evaluation of the data provides a feedback loop that enables assessment of the Designs relative to Observatory goals. Moreover, results of these analyses will allow the NEON to effectively prioritize sampling costs in the face of uncertain future funding levels.

Specification of analyses suitable for evaluating the efficacy of the NEON sampling designs required development of specific questions. Power analyses and sampling simulations were then used to determine the degree to which design modifications affected the ability of the data to address those questions. Questions and analytical approaches were developed in consultation with NEON Technical Working Groups comprised of subject matter experts selected from the ecological community.

The two primary and actionable questions of a 2018 effort for these data were:

1. Where is the variability in the data? Specifically, what proportion of the variance at each site exists at the year, bout, plot, and subplot scales? Understanding the components of variance provides a critical component of the actionable directions for modifying sampling size, subplot sampling intensity, and temporal sampling frequency.
2. Is it possible to detect meaningful interannual variation in key response variables? Understanding the power to detect change between years provides a direct indication of the sample sizes required to detect meaningful ecological change.

The variance components analysis supported the reduction in the number of the number nested 10m² and 1m² subplots within each plot from eight to six. Additionally, the 10m² and 100m² subplots at which species presence are recorded were reduced to every other year sampling. The 1m² subplots at each plot continue to be sampled annually.

The power analysis reduced the total number of plots sampled at a small number of sites. Only sites with three or more years of data were considered. An initial goal was to evaluate trend detection, but it was decided insufficient years of data had been collected for that analysis.

With many more years of data available across all NEON terrestrial sites in 2026 (Appendix A), this effort revisits the question of detectability of interannual change in plant community composition and abundance. Specifically, we ask:

1. What level of spatial replication (number of plots) is required to reliably detect a specified relative change in plant species cover at a site?
2. What level of spatial replication is required for reliable estimation of multi-year directional trends in species cover?

Together, these analyses provide complementary bounds on the minimum sampling effort required to detect meaningful ecological change at short (interannual) and long (multi-year trend) temporal horizons.

2 RELATED DOCUMENTS AND ACRONYMS

2.1 Reference Documents

RD[01]	NEON.DOC.000001	NEON Observatory Design
RD[02]	NEON.DOC.002652	NEON Level 1, Level 2, and Level 3 Data Products Catalog
RD[03]	NEON.DOC.000912	TOS Science Design for Plant Diversity
RD[04]	NEON.DOC.014042	TOS Protocol and Procedure: Plant Diversity Sampling

2.2 Acronyms

Acronym	Definition
NEON	National Ecological Observatory Network
TOS	Terrestrial Observation System
TWG	Technical Working Group
NLCD	National Land Cover Database
GJAM	Generalized Joint Attribute Model
CA	Censored Abundance (GJAM likelihood for cover data)
MCMC	Markov Chain Monte Carlo

3 METHODS

We evaluated the capacity of NEON’s terrestrial plant sampling design to detect ecological change in species- and community-level response variables using a hierarchical Bayesian modeling framework combined with simulation-based power and sensitivity analyses. The assessment comprises two complementary analyses:

1. We conducted a perturbation-based power analysis to quantify the spatial replication (number of randomly placed plots) required to detect a specified magnitude of interannual change in species cover, evaluated through a paired counterfactual design that controls for natural year-to-year variability.
2. We evaluated the spatial replication required for reliable estimation of multi-year directional trends, including the stability of trend estimates across sample sizes and the detectability of imposed trends of known magnitude.

Together, these analyses address the two primary temporal scales at which change detection is relevant to NEON’s mission: abrupt interannual shifts and gradual directional trends.

3.1 Data and preprocessing

The observation unit for modeling is a species-specific cover value at the plot-year level. Within each plot-year, the six (or eight prior to 2019) 1m² subplot observations are averaged to produce a single cover value per species per plot per year. For example, at a site where 30 plots are sampled annually for 8

years, a species recorded on all plots would contribute 240 plot-year observations to the model. The full model input for a site is the matrix of all species $\times$ all plot-years, with cover values as entries.

Prior to model fitting, species occurring in fewer than three plots across the entire site record are aggregated into a composite “OTHER” category. This filter removes species for which spatial replication is insufficient to estimate covariance structure, while retaining them in the modeled community so that their aggregate contribution to total cover is represented (Clark et al. 2017). Species with all-zero cover or zero variance after aggregation are removed. The “OTHER” category participates in model fitting but is excluded from power and trend analyses, which focus on individually identifiable species.

Separately, a cover floor of 3% mean raw field cover is applied when evaluating power and trend stability. Species below this threshold are excluded from the power analysis because a 20% relative change at 3% cover corresponds to 0.6 percentage points—below the 1% recording resolution of NEON’s field protocol. This cover floor is computed from the raw field observations (mean of percent cover per species across all plot-years at the site) and is independent of the GJAM model predictions which undergo compositional normalization that compresses cover values.

3.2 Joint species distribution model

Both analyses share a common modeling framework. Plant species cover was modeled with generalized joint attribute models (GJAM; Clark et al. 2017), a multivariate hierarchical Bayesian framework that jointly represents all species at a site while accounting for shared responses to environmental covariates and residual covariance among species. A single GJAM model is fit per site using all available plot-years. Both the perturbation-based and trend analyses draw on the posterior output of this shared fit. For plot p in year t , the vector of observed percent cover values for all S species is modeled as:

$$\mathbf{y}_{p,t} \sim \text{CA}(\mu_{p,t}, \Sigma)$$

$$\mu_{p,t} = \mathbf{X}_{p,t} \mathbf{B}$$

where $\mathbf{y}_{p,t}$ is the S -vector of observed percent cover, $\mu_{p,t}$ is the latent mean cover vector, $\mathbf{X}_{p,t}$ is the design matrix of covariates, $\mathbf{B}$ is the $K \times S$ matrix of species-specific regression coefficients (equivalent to the transpose of the coefficient matrix in Clark et al. 2017), and Σ represents residual covariance among species not explained by the covariates, where K is the number of covariates.

The censored-abundance (CA) likelihood represents observed cover as a censored realization of an underlying multivariate normal latent variable, allowing exact zeros as well as continuous positive values.

Two model specifications were used depending on land cover heterogeneity at each site: $\mathbf{Y} \sim \text{year} + \text{nlcdClass}$ at sites with multiple National Land Cover Database (NLCD) classes represented among sample plots, and $\mathbf{Y} \sim \text{year}$ at sites with a single class or where the multi-class specification failed to converge. Year was treated as a categorical factor. The MCMC sampler was run for 5,000 iterations; 2,500 iterations were discarded as burn-in, yielding 2,500 retained posterior draws. For downstream posterior prediction and sensitivity analyses, we used $G = 1,000$ posterior draws sampled from the retained MCMC output for computational efficiency.

3.2.1 Posterior prediction and dimension reduction

Joint modeling of plant communities at NEON sites frequently involves hundreds of species observed across tens of plots (Appendix here to numbers of sp), producing covariance matrices that are high-dimensional relative to the available sample size. To illustrate, consider a southeastern forest site with 400 plant species observed across 30 plots per year: estimating the 400×400 residual covariance matrix Σ from 300 plot-year observations is not feasible using standard approaches.

GJAM addresses this challenge through a Bayesian dimension reduction framework (Taylor-Rodriguez et al. 2017; Clark et al. 2017). Rather than estimating Σ directly, the model represents the S -dimensional species response through a lower-dimensional factor structure. Species are assigned to N_c clusters via a truncated Dirichlet process representation, and within each cluster, an r -dimensional factor loading matrix $\mathbf{Z}$ captures shared covariance. The implied covariance is:

$$\Sigma = \mathbf{A}\mathbf{A}' + \sigma_\varepsilon^2 \mathbf{I}$$

where $\mathbf{A} = \mathbf{Q}(\mathbf{k})\mathbf{Z}$ is the $S \times r$ loading matrix that maps the reduced factor structure back to the full species space via cluster assignments $\mathbf{k}$, and σ_ε^2 is a scalar residual variance. This factorization reduces the effective number of covariance parameters from $S(S+1)/2$ to $(N_c \times r) + 1$, enabling stable estimation even when S greatly exceeds the number of observations n .

GJAM invokes dimension reduction when the number of species is large relative to the number of observations, which was the case for all NEON sites analyzed here (Taylor-Rodriguez et al. 2017). This condition was met at all 46 NEON sites, where species counts ranged from 39 to 406 while plot-year observations ranged from approximately 100 to 400. We therefore set reduction explicitly with a truncation level of $N_c = 20$ clusters and $r = 5$ factors across all sites, ensuring consistent parameterization and comparability of posterior predictions across the Observatory. Posterior predictive draws were reconstructed from the stored MCMC chains for $\mathbf{B}$, $\mathbf{k}$, $\mathbf{Z}$, and σ_ε^2 using the reduced-rank generative model:

$$\mathbf{y}_i^{(g)} = \mathbf{x}_i \mathbf{B}^{(g)} + \mathbf{w}_i^{(g)} \mathbf{A}^{(g)'} + \varepsilon_i^{(g)}$$

where $\mathbf{x}_i \mathbf{B}^{(g)} = \mu_i^{(g)}$ is the latent mean response defined above, $\mathbf{B}^{(g)}$ is the $K \times S$ matrix of species-specific regression coefficients for posterior draw g , $\mathbf{A}^{(g)}$ is the $S \times r$ loading matrix reconstructed from cluster assignments $\mathbf{k}^{(g)}$ and factor loadings $\mathbf{Z}^{(g)}$ at draw g , $\mathbf{w}_i^{(g)}$ is an r -dimensional latent factor score with $\mathbf{w}_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$, and $\varepsilon_i^{(g)}$ is species-level residual error with $\varepsilon_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^{2(g)} \mathbf{I})$. Superscript (g) indexes the posterior draw used for prediction ($g = 1, \dots, G$, with $G = 1,000$ in the sensitivity analyses). For each draw, the parameters $\mathbf{B}^{(g)}$, $\mathbf{A}^{(g)}$, and $\sigma_\varepsilon^{2(g)}$ are fixed, while $\mathbf{w}_i^{(g)}$ and $\varepsilon_i^{(g)}$ are drawn independently, yielding posterior predictive realizations that preserve the covariance structure implied by the reduced-rank model without explicitly forming the full $S \times S$ covariance matrix.

The analysis propagates key sources of uncertainty through the posterior predictive distribution. Parameter uncertainty is captured through variation across MCMC draws, while residual variability—representing ecological and measurement variation not explained by covariates—is captured through the stochastic

components $\mathbf{w}$ and ε . Observation uncertainty (field measurement error) is not modeled explicitly but is implicitly absorbed into the residual variance estimates.

3.3 Year-to-year change detection (counterfactual analysis)

This analysis was designed to evaluate the detectability of year-to-year changes in plant species cover, analogous to earlier frequentist approaches that compare observed changes between consecutive years. However, because natural interannual variability can obscure imposed signals, we instead use a counterfactual design in which both baseline and perturbed predictions are generated from the same year. This design isolates statistical detectability from ecological variability among years, providing a controlled evaluation of monitoring performance under realistic covariance structure. For each site and for each year with at least five plots sampled, paired counterfactual prediction sets were generated: a baseline set and a perturbed set, both derived from the same year's data using the same posterior parameter draws but with independently sampled observation-level noise realizations ($\mathbf{w}$ and ε). This paired counterfactual design isolates the imposed perturbation from natural year-to-year variation, ensuring that detected differences are attributable to the imposed effect rather than to ecological differences between years.

Let $\mu_{s,p}^{(g)}$ denote the posterior predictive cover for species s at plot p for MCMC draw g .

The perturbation was applied directly to the posterior predictive draws as a per-species additive shift proportional to each species' mean raw field cover:

$$\mu_{s,p}^{\text{pert},(g)} = \mu_{s,p}^{(g)} + \delta \cdot \bar{c}_s$$

where $\mu_{s,p}^{(g)}$ is the posterior predictive cover for species s at plot p for draw g , $\mu_{s,p}^{\text{pert},(g)}$ is the corresponding perturbed value, $\bar{c}_s$ is the mean percent cover of species s computed from raw field observations across all plot-years at the site, and δ is the relative effect size. A species at 10% mean cover with $\delta = 0.20$ receives a +2 percentage point perturbation; a species at 60% mean cover receives +12 percentage points. This per-species scaling ensures every species faces the same proportional challenge regardless of baseline cover level. We tested effect sizes of $\delta \in \{0.15, 0.20, 0.25, 0.30\}$, corresponding to 15–30% relative changes in cover.

For each site and year, the baseline and perturbed prediction sets were subsampled to represent alternative levels of spatial replication. All plots sampled in a given year were eligible for subsampling; no requirement was imposed that plots be sampled in consecutive years, as the paired counterfactual design generates both prediction sets from the same year's data. Subsamples consisted of K plots drawn without replacement, with $K \in \{30, 25, 20, 15, 10, 8, 6, 5\}$, and each subsampling was repeated 50 times.

For each Monte Carlo replicate, the site-level mean cover for species s was computed for both the baseline and perturbed prediction sets by averaging across the K sampled plots within each posterior draw g :

$$\bar{\mu}_{s,K}^{(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p}^{(g)}, \quad \bar{\mu}_{s,K}^{\text{pert},(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p}^{\text{pert},(g)}$$

where $\mu_{s,p}^{(g)}$ and $\mu_{s,p}^{\text{pert},(g)}$ denote the baseline and perturbed posterior predictive cover values, respectively.

Detection was evaluated using a per-species absolute difference criterion. For each species s , detection probability was defined as:

$$P\left(\left|\bar{\mu}_{s,K}^{\text{pert},(g)} - \bar{\mu}_{s,K}^{\text{base},(g)}\right| \geq \tau_s\right) \geq 0.80$$

where $\bar{\mu}_{s,K}^{\text{base},(g)}$ and $\bar{\mu}_{s,K}^{\text{pert},(g)}$ are the site-level mean cover values computed from the baseline and perturbed prediction sets, respectively, for posterior draw g , and τ_s is the species-specific detection threshold. The probability is computed as the fraction of posterior draws g for which the absolute difference exceeds τ_s . The ratio of imposed effect to detection threshold is constant across species (2:1 at $\delta = 0.20$), providing equivalent statistical difficulty regardless of cover level. This ensures that detectability comparisons are not driven by differences in species abundance scale.

A species was classified as detectable at sample size K and effect δ when the above criterion was satisfied in at least 80% of the 50 Monte Carlo replicates (power ≥ 0.80). The community-level metric reports the percentage of species (above the 3% cover floor) that are individually detectable at each K .

3.4 Multi-year trend analysis

3.4.1 Objective and conceptual framing

The counterfactual change detection analysis evaluates the ability to detect a discrete change at a single time point. However, many ecological changes of interest to NEON manifest as gradual directional trends that accumulate over multiple years. A site might experience a slow decline in native grass cover or a progressive increase in woody encroachment that is undetectable in any single year-to-year comparison but becomes apparent over a longer observation window. The trend analysis addresses this complementary question: what level of spatial replication is required for reliable estimation and detection of directional trends in species cover across a multi-year observation window?

This question was identified as a priority by the NEON Technical Working Group during the 2018 design optimization effort (Barnett et al. 2018) but could not be addressed at that time due to limited temporal replication. With 8–12 years of data now available at most NEON sites, the question becomes tractable.

3.4.2 Moving-window framework

Trend detection was evaluated using a moving-window approach (Urquhart et al. 1998). Rather than fitting a single trend line through the entire time series, we applied overlapping windows of fixed width W that slide across the available years. This approach reflects realistic monitoring practice: an analyst examining NEON data in any given year would assess the most recent W years to determine whether a trend is present. It also accommodates the possibility that trend strength varies across the observation period.

For each site, we evaluated window widths of $W \in \{5, 10\}$ years. Within each window, a trend was quantified as the ordinary least-squares slope of species cover against year, computed from posterior pre-

dictive site-year means derived from plot-level predictions. This yields a posterior distribution of slope estimates for each species, window position, and sample size.

3.4.3 Observed trend detection

The observed trend detectability analysis evaluates whether trends already present in the posterior predictions are detectable as sample size changes. For each species, window position, posterior draw, and sample size K , the site-level mean cover was computed by averaging across the K sampled plots within each year:

$$\bar{\mu}_{s,t,K}^{(g)} = \frac{1}{K} \sum_{p=1}^K \mu_{s,p,t}^{(g)}$$

where $\mu_{s,p,t}^{(g)}$ is the posterior predictive cover for species s at plot p in year t for MCMC draw g . A temporal trend was then estimated for each species and posterior draw as the ordinary least-squares slope of $\bar{\mu}_{s,t,K}^{(g)}$ against centered year:

$$\gamma_{s,K}^{(g)} = \frac{\sum_t (t - \bar{t}) \bar{\mu}_{s,t,K}^{(g)}}{\sum_t (t - \bar{t})^2}$$

Detection was defined as the posterior probability that the absolute slope exceeded a biologically meaningful threshold:

$$P\left(|\gamma_{s,K}^{(g)}| \geq \frac{\delta}{T}\right) \geq 0.80$$

where $\gamma_{s,K}^{(g)}$ is the estimated trend slope for species s at sample size K , $\delta = 0.20$ corresponds to a cumulative 20% change over the window, and $T = t_{\max} - t_{\min}$ of the window, defined as the final year minus the first year. Plots were sampled independently within each year of the window, allowing different plots to contribute to the trend estimate in different years.

Observed trend detectability answers the question: given the trends already present at a site, how many plots would be needed to detect them with high probability?

3.4.4 Imposed trend detection

The imposed trend detectability analysis evaluates the ability to detect a trend of known magnitude that may not be present in the observed data. This provides a prospective analysis analogous to the counterfactual change detection analysis, but for gradual directional trends rather than discrete changes at a single time point. For each window, an additive linear trend was imposed on the posterior predictive means before estimating slopes:

$$\mu_{s,p,t}^{\text{trend},(g)} = \mu_{s,p,t}^{(g)} + \frac{\delta}{T}(t - t_0)$$

where $\mu_{s,p,t}^{(g)}$ is the original posterior predictive cover, $\mu_{s,p,t}^{\text{trend},(g)}$ is the prediction after imposing the trend, t_0 is the first year of the window, δ is the imposed cumulative change, and T is the calendar time span of the window. Slopes $\gamma_{s,K}^{(g)}$ were then estimated from the trend-modified predictions using the same centered-year regression described above. Detection was assessed using the same posterior probability criterion as in the observed trend analysis.

Imposed trend detectability answers the question: if a trend of the specified magnitude were to emerge at a site, how many plots would be required to detect it?

3.4.5 Slope estimation stability

The detection-based analyses above evaluate whether an estimated slope exceeds a fixed threshold. A complementary question is whether the slope estimate itself is stable—that is, whether it would yield a similar value if a different set of plots were selected. This is directly relevant to monitoring design: a slope estimate that varies substantially depending on which plots are sampled provides a weak basis for inference, regardless of whether it exceeds a detection threshold.

To evaluate this, we computed the coefficient of variation (CV) of the slope estimate across Monte Carlo replicates at each sample size K . For each species s , window position, and sample size, each of the $R = 50$ Monte Carlo replicates yields a mean slope (averaged across posterior draws):

$$\bar{\gamma}_{s,K,m} = \frac{1}{G} \sum_{g=1}^G \gamma_{s,K,m}^{(g)}$$

where $\gamma_{s,K,m}^{(g)}$ is the slope estimate for species s at sample size K in Monte Carlo replicate m for posterior draw g , and G is the number of posterior draws.

The coefficient of variation across replicates is then:

$$\text{CV}_s(K) = \frac{\text{SD}(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R})}{|\text{mean}(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R})|}$$

At low K , slope estimates vary widely depending on which plots are selected (high CV); as K increases, estimates converge (low CV). A CV below 0.50 indicates that the standard deviation of the slope estimate is less than half its magnitude, such that variation in plot selection does not qualitatively change the inferred direction and strength of the trend.

This stability analysis provides a perspective complementary to the detection-based analyses. Adding spatially heterogeneous plots to annual site means can increase year-to-year variability, making threshold-based detection more difficult at higher K —an effect that reflects real ecological heterogeneity but com-

plicates interpretation of detection curves. In contrast, the CV metric evaluates the precision of the slope estimate itself and decreases monotonically with K at nearly all sites.

For each species and site, we defined K_{CV50} as the smallest sample size at which $CV_s(K) \leq 0.50$, representing the point at which slope estimates become stable with respect to plot selection.

3.4.6 Definition of N^* for trend detection

For each site, species, and window width, we defined N^* as the smallest number of plots K for which the trend detection criterion was met. As with the counterfactual change detection analysis, N^* values were computed separately for each species and then summarized at the community level.

3.5 Scope and relationship between analyses

The counterfactual year-to-year change detection and trend analyses address ecological change at different temporal scales. The counterfactual year-to-year change detection analysis quantifies the spatial replication required to detect a specified relative change at a single time point, providing guidance relevant to monitoring for abrupt ecological shifts. The trend analysis evaluates the spatial replication required for reliable trend estimation over multi-year windows, providing guidance relevant to tracking gradual directional changes.

These analyses are complementary. The slope stability analysis from the trend framework directly parallels the counterfactual change detection curves: both ask how the quality of inference improves with additional plots. Where both analyses converge on similar sample size requirements, the recommendation is strengthened.

4 RESULTS

We present results from site-level analyses across the NEON network, drawing on both the counterfactual change detection framework and the multi-year trend analysis. Together, these analyses evaluate how the ability to detect ecological change and estimate directional trends depends on spatial replication (number of plots) under realistic levels of spatial and temporal variability.

The analyses leverage 8–12 years of NEON vegetation data per site, representing a substantial increase in temporal replication relative to the 2018 design optimization effort. This expanded record enables direct evaluation of both year-to-year change detection and multi-year trend estimation within a unified modeling framework.

We begin with a summary of site-level data characteristics, including the number of years of observation and the number of species retained after applying the cover filter. We then present detailed results for a small number of representative sites to illustrate the behavior of each analysis, followed by summary tables that synthesize results across all completed sites.

Across completed sites, the number of years of observation ranged from X to Y, with most sites having at least eight years of data. The number of species included in the GJAM model ranged from X to Y per site, reflecting substantial variation in community complexity across the network. After applying the 3% mean cover threshold, the number of species retained for analysis was reduced to X–Y per site. This

filtering removes species for which a 20% relative change falls within the resolution of the field protocol, ensuring that detection and stability metrics are evaluated on the portion of the community where meaningful change can be resolved. Species that persist through that filtering were included in model fitting and posterior prediction. Detection and stability metrics were restricted to species with mean raw cover $\geq 3\%$, and this same filtered species set was used consistently across both the change detection and trend analyses. The number of years of observation and the number of species represented at different stages of the workflow vary by site. For each site, we report three species counts: (1) the total number of species observed in the raw dataset, (2) the number of species included in the model, and (3) the number of species retained for inference after applying the 3% mean cover threshold (Table 1).

Table 1: Summary of site characteristics, including years of observation and species counts for observed, modeled, and inference subsets.

Domain	Site	Site Code	Year Range	Species (observed)	Species (modeled)	Species ($\geq 3\%$ cover)
D16	Abby Road	ABBY	8 2017-2024	247	129	43
D18	Utqiagvik	BARR	7 2017-2024	90	56	10
D01	Bartlett Experimental Forest	BART	11 2014-2024	95	44	7
D02	Blandy Experimental Farm	BLAN	10 2015-2024	292	149	30
D19	Caribou-Poker Creeks Research Watershed	BONA	8 2017-2024	84	46	26
D11	Lyndon B. Johnson National Grassland	CLBJ	7 2016-2024	491	249	63
D10	Central Plains Experimental Range	CPER	12 2013-2024	206	122	22
D09	Dakota Coteau Field Site	DCFS	8 2017-2024	264	143	36
D19	Delta Junction	DEJU	9 2016-2024	182	98	24
D03	Disney Wilderness Preserve	DSNY	10 2014-2024	612	304	92
D07	Great Smoky Mountains National Park	GRSM	5 2015-2024	259	126	47
D04	Guánica Forest	GUAN	10 2015-2024	229	121	77
D01	Harvard Forest & Quabbin Watershed	HARV	11 2014-2024	322	145	47

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D19	Healy	HEAL	10	2015-2024	136	78	36
D03	The Jones Center At Ichauway	JERC	10	2014-2024	823	406	122
D14	Jornada Experimental Range	JORN	9	2015-2024	171	91	5
D06	Konza Prairie Agroecosystem	KONA	8	2017-2024	113	46	9
D06	Konza Prairie Biological Station	KONZ	5	2016-2024	355	195	52
D04	Lajas Experimental Station	LAJA	9	2016-2024	226	108	83
D08	Lenoir Landing	LENO	8	2016-2024	289	161	33
D07	Mountain Lake Biological Station	MLBS	8	2017-2024	217	124	48
D13	Moab	MOAB	8	2015-2024	174	95	4
D13	Niwot Ridge	NIWO	10	2015-2024	247	147	30
D09	Northern Great Plains Research Laboratory	NOGP	9	2016-2024	257	125	49
D11	Marvin Klemme Range Research Station	OAES	9	2015-2024	397	224	47
D15	Onaqui	ONAQ	10	2014-2024	181	90	9
D07	Oak Ridge	ORNL	11	2014-2024	370	186	64
D03	Ordway-Swisher Biological Station	OSBS	11	2013-2024	512	212	54
D20	Pu'u Maka'ala Natural Area Reserve	PUUM	7	2018-2024	122	61	20
D10	Rocky Mountains	RMNP	8	2017-2024	215	98	19
D02	Smithsonian Conservation Biology Institute	SCBI	11	2014-2024	406	226	58
D02	Smithsonian Environmental Research Center	SERC	9	2015-2024	285	171	35

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species ($\geq 3\%$ cover)
D17	San Joaquin Experimental Range	SJER	8	2017-2024	252	130	40
D17	Soaproot Saddle	SOAP	6	2018-2024	317	131	46
D14	Santa Rita Experimental Range	SRER	9	2016-2024	441	271	38
D05	Steigerwaldt-Chequamegon	STEI	10	2015-2024	418	190	71
D10	North Sterling	STER	11	2014-2024	70	39	16
D08	Talladega National Forest	TALL	11	2014-2024	452	244	64
D17	Lower Teakettle	TEAK	5	2019-2024	199	70	29
D18	Toolik Field Station	TOOL	8	2017-2024	188	105	19
D05	Treehaven	TREE	10	2015-2024	411	191	49
D06	University of Kansas Field Station	UKFS	9	2016-2024	383	207	68
D05	University of Notre Dame Environmental Research Center	UNDE	11	2014-2024	393	198	44
D09	Chase Lake National Wildlife Refuge	WOOD	11	2014-2024	282	127	45
D16	Wind River Experimental Forest	WREF	7	2018-2024	123	60	23
D12	Yellowstone National Park	YELL	6	2018-2024	421	185	29

4.1 Perturbation-based power analysis

4.1.1 Community-level power curves

At all four validation sites, community-level detection power increased monotonically with sample size at the focal effect size of $\delta = 0.20$ (20% relative change). After applying the 3% cover floor, the number of species included in the community metric ranged from 45 (HARV) to 55 (OSBS).

At $\delta = 0.20$, all four sites crossed the 80% community detection threshold between $K = 20$ and $K = 25$ plots: HARV at $K = 25$ (80%), OSBS at $K = 20$ (83%), KONZ at $K = 25$ (80%), and GRSM at $K =$

20 (82%). At $\delta = 0.15$ (15% relative change), detection remained below 45% even at $K = 30$, indicating that changes below 20% are not reliably detectable at any sample size under the current framework. At $\delta = 0.25$, detection was nearly saturated ($>90\%$) at all sample sizes including $K = 5$.

4.1.2 Year-to-year stability

Detection power was stable across baseline years within each site. At $\delta = 0.20$ and $K = 20$, the standard deviation of community detection across years was 0–2.4 percentage points (KONZ: 0%, HARV: 1.8%, GRSM: 2.2%, OSBS: 2.4%). This stability confirms that the paired counterfactual design successfully eliminates year-to-year confounds.

4.2 Multi-year trend analysis

4.2.1 Slope estimation stability (primary result)

The slope stability analysis showed that trend estimates became progressively more reliable as sample size increased. Across the network, the median coefficient of variation (CV) of species-level slope estimates decreased monotonically with K at virtually all sites for both 5-year and 10-year windows. The number of plots required for stable trend estimation (median $CV < 0.50$) ranged from 5 at spatially homogeneous sites (e.g., arid grasslands) to 20 or more at topographically complex sites (e.g., montane forests), providing genuine site-level differentiation.

These results complement the perturbation-based findings: both analyses identify $K = 20$ –25 as the sample size at which inference quality transitions from unreliable to adequate at most sites, albeit through different metrics (detection power vs. estimation precision).

4.2.2 Imposed trend detectability (supplementary)

The detection-based trend analyses revealed a spatial dilution effect at a number of sites: detection probability decreased rather than increased with additional plots. This occurs because adding spatially heterogeneous plots increases the year-to-year scatter of annual site means, making OLS slopes harder to distinguish from zero against a fixed threshold. This effect is ecologically real—more plots sample more habitat variation—and represents a fundamental difference between trend detection and point-in-time change detection. Detection-based trend power curves should therefore be interpreted with this caveat.

4.2.3 Plot continuity and the COVID-2020 gap

The COVID-2020 sampling disruption reduced the number of plots sampled at many sites, creating continuity gaps that affect multi-year trend analysis. To address this, the trend analysis samples plots independently for each year within a window, rather than requiring the same plots to be sampled in every year. This design reflects realistic monitoring conditions where plot availability may vary across years.

5 DISCUSSION

5.0.1 Complementary perspectives on sampling adequacy

This analysis demonstrates that conclusions about required spatial replication depend on both the magnitude of ecological change and the temporal scale over which it is assessed. The perturbation-based power analysis quantifies the ability to detect a specified relative change in species cover at a single time point, providing guidance relevant to monitoring for abrupt ecological shifts—invasive species establishment, disturbance events, or rapid phenological changes. The trend stability analysis quantifies the number of plots required for reliable estimation of multi-year directional trends, providing guidance relevant to tracking gradual changes in community composition driven by climate, land use, or successional dynamics.

Both analyses converge on a consistent recommendation: 20–25 plots are required for reliable ecological inference. The perturbation analysis identifies this range as the sample size at which 80% of species (above the 3% cover floor) are detectable at a 20% relative change. The trend stability analysis identifies this range as the sample size at which slope estimates become stable ($CV < 0.50$ for most species). The convergence of these independent analyses—addressing different temporal scales through different statistical frameworks—strengthens confidence in the recommendation.

5.0.2 Per-species relative framework

The use of per-species perturbation and detection thresholds scaled to each species' raw mean cover represents an important methodological choice. By imposing a 20% relative change on every species (rather than a fixed absolute change), the analysis ensures that low-cover and high-cover species face equivalent proportional challenges. A species at 10% mean cover receives a 2 percentage point perturbation against a 1 percentage point threshold; a species at 60% cover receives 12 percentage points against 6. This scaling avoids the bias inherent in fixed absolute frameworks, where a given percentage-point change is ecologically trivial for dominant species but dramatic for subordinate ones.

The 3% cover floor reflects the practical limit of both field measurement precision (NEON records cover to the nearest 1%) and model prediction reliability at low cover. Below this threshold, a 20% relative change falls within the recording resolution of the field protocol. At most NEON sites, the majority of species fall below the 3% floor, meaning that the power and stability analyses characterize detection capability for the more abundant portion of the community.

5.0.3 Modeling framework and dimension reduction

The use of GJAM as the underlying modeling framework enables joint inference across all species at a site, accounting for shared environmental responses and residual covariance among species. This represents a substantial methodological advance over the univariate mixed-effects models employed in the 2018 optimization analysis (Barnett et al. 2018), which treated each species independently and could not account for the coordinated responses that characterize plant communities.

The dimension reduction framework (Taylor-Rodriguez et al. 2017) was essential for applying this approach across the full diversity gradient of NEON sites. Without reduction, the $S \times S$ covariance matrix cannot be estimated reliably when the number of species exceeds the effective sample size, a condition

met at the majority of NEON sites. By representing the covariance through a reduced factor structure, the model captures the dominant modes of inter-species association while remaining computationally tractable at sites with hundreds of species.

5.0.4 Spatial dilution in trend detection

The detection-based trend analyses revealed an important finding: at many sites, the probability of detecting a trend against a fixed threshold decreased as more plots were added. This spatial dilution effect occurs because additional plots sample more habitat variation, increasing the year-to-year scatter of annual site means and making OLS slopes harder to distinguish from zero. This effect is ecologically real and represents a fundamental difference between trend detection against a fixed threshold and point-in-time change detection, where averaging across more plots always reduces noise.

The slope stability analysis was developed to complement the detection-based approach. Because stability measures the precision of the slope estimate (via the coefficient of variation across plot resamples) rather than its exceedance of a threshold, it does not suffer from the spatial dilution artifact. More plots always provide a more representative sample of the site, reducing the sensitivity of the trend estimate to which specific plots were selected.

5.0.5 Implications for NEON sampling design

The results indicate that NEON's current design of 20–30 plots per site is well-calibrated for detecting 20% relative changes in plant species cover and for producing stable trend estimates. At four validation sites spanning temperate deciduous forest, tallgrass prairie, montane forest, and subtropical pine flatwoods, all sites crossed the 80% community detection threshold at $K = 20$ –25 for a 20% relative effect. Across the full network, most sites showed consistent patterns, with site-level variation in trend stability providing genuine differentiation: spatially homogeneous sites (arid grasslands, agricultural systems) achieve stable trend estimates with as few as 5 plots, while topographically complex sites require 15–20 or more.

Changes below 15% relative are not reliably detectable at any sample size under the current framework, establishing a practical floor on the ecological sensitivity of NEON's plot-based monitoring. Changes of 25% or larger are detectable with as few as 5 plots, indicating that major ecological events will be captured regardless of sampling intensity.

5.0.6 Addressing the 2018 trend detection question

The 2018 design optimization effort identified trend detection as a priority question but determined that it could not be addressed with the limited temporal replication then available (Barnett et al. 2018). With 8–12 years of data now available at most sites, the moving-window trend analysis presented here directly addresses that outstanding question. The slope stability framework provides actionable guidance: 20–25 plots are sufficient for reliable trend estimation over 5–10 year windows at most sites, consistent with the perturbation-based change detection results.

5.0.7 Limitations and future directions

Several limitations of the current analysis warrant emphasis. First, the model is fit once per site using all available data, and posterior predictions are generated conditional on this single fit (single-fit design). The analysis does not propagate uncertainty arising from plot selection into the model fitting step, meaning that power estimates represent upper bounds on detection capability. The consistency of recommendations across ecologically distinct sites and with the 2018 univariate analysis provides confidence that the directional findings—20–25 plots for reliable change detection and trend estimation—are robust, though the true sample size requirement may be somewhat higher. A limited refit analysis, in which the model is re-estimated at each sample size, could quantify this optimism and is identified as a priority for future work.

Second, multiple relative effect sizes (15–30%) were tested, revealing that 20% is the critical threshold where sample size determines detection capability. Sensitivity to alternative detection threshold values (currently 10% of raw cover) was not systematically evaluated but could be addressed in future work.

Third, the GJAM-based approach treats observed cover values as compositional, normalizing species shares within each observation. This compositional normalization compresses absolute cover values relative to raw field data, which is why the cover floor and per-species scaling are computed from raw field observations rather than model predictions.

The analytical framework developed here is extensible in several directions. Longer NEON time series will enable evaluation of trend detection over wider windows and provide empirical validation of the power estimates. The GJAM-based approach could also be extended to other NEON observation systems where multivariate community data are collected, including small mammals, ground beetles, and soil microorganisms.

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